

PSRP Executive Controlled Review

A curated executive readout of the PSRP web surface. This page does not mirror the application section by section; it explains the professional argument behind the prototype: local-first authority, regional interpretation, bounded analytics, and publication-safe claims.

EXECUTIVE READOUT

What a reviewer should understand first

PSRP is strongest when read as an engineering artifact, not as a map demo. Its value is the controlled path from curated evidence to SQLite authority, derived regional context, bounded model posture, and a public surface that avoids unsupported warning-system claims.

ARCHITECTURE

local-first prototype with SQLite authority

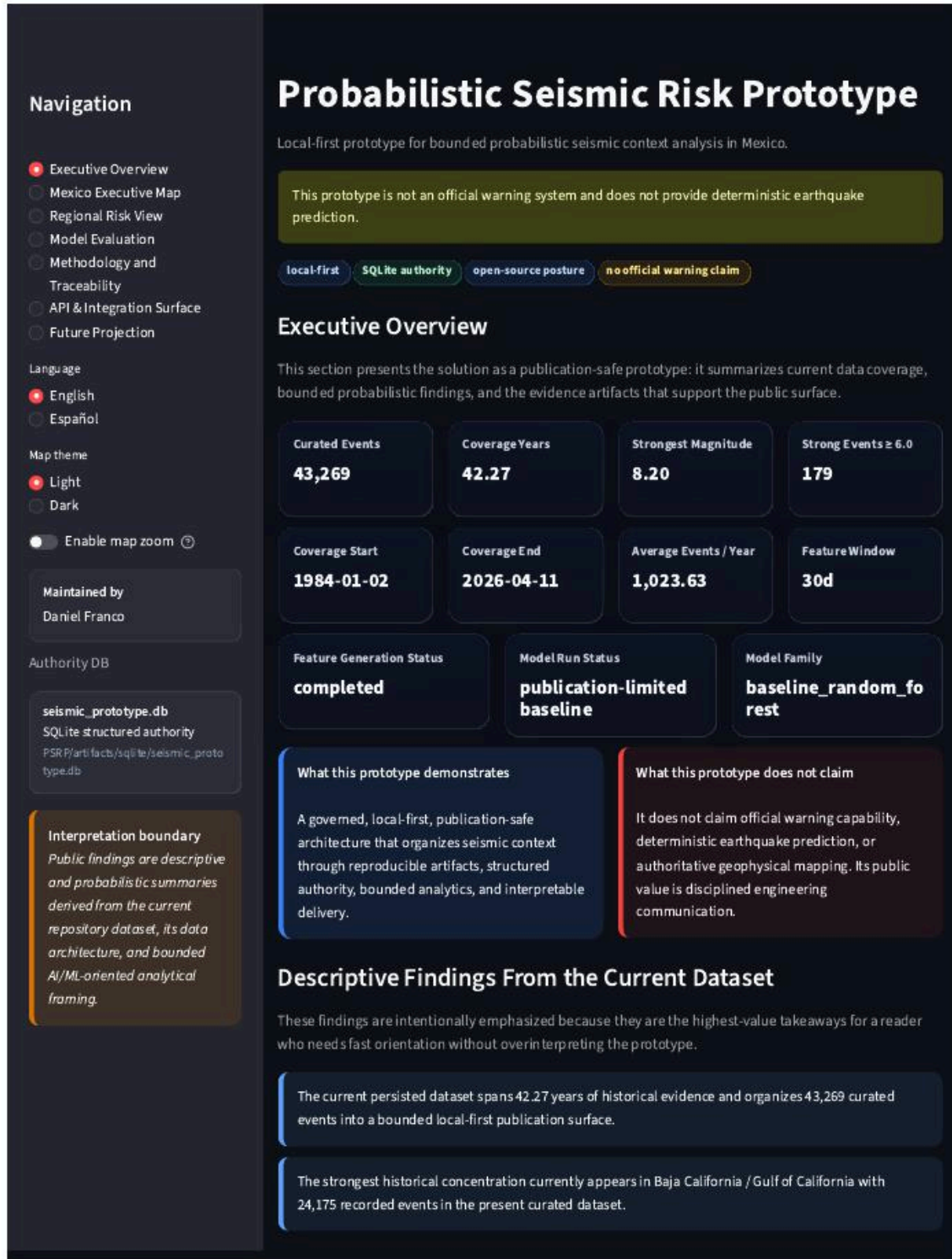
ANALYTICS

probabilistic regional context, not prediction

GOVERNANCE

bounded claims and reproducible artifacts

Publication signal: the visible evidence surface reports 43,269 curated events, 1984-01-02 to 2026-04-11 coverage, strongest magnitude 8.20, a 30-day feature window, and a publication-limited baseline model posture.



Current entry surface used as a posture reference: scale, coverage, model status, and non-claim boundary are visible before the reader explores details.

Executive readout

Technical authority

Map interpretation

Regional reading

Model posture

Integration

Executive conclusion

For recruiters

Read this as: a compact proof that the portfolio artifact is polished, bounded, and professionally communicated without exaggerated claims.

For engineering reviewers

Read this as: an architecture surface where raw evidence, structured authority, derived features, evaluation, and publication are deliberately separated.

For publication review

Read this as: an evidence-based interpretation layer that connects current dataset scale, technical authority, bounded model posture, and explicit non-warning semantics.

01 / TECHNICAL AUTHORITY

Traceability is the core credibility mechanism

Methodology & Traceability Cycle

Local-first lineage from immutable evidence to bounded publication surface — not an official warning system.



The most important professional argument is not the map alone. It is the controlled lineage from source evidence to structured authority, derived regional features, bounded evaluation, and publication-safe communication.

Acceptance signal: the methodology should read like auditable engineering, not over-documentation.

LAYER	EXECUTIVE MEANING	CREDIBILITY CONTRIBUTION
Raw evidence	Source material remains distinguishable from derived outputs.	Supports auditability and safe reruns.
SQLite authority	Structured local state becomes the prototype source of truth.	Reduces ambiguity and avoids cloud dependency.
Feature / region layer	Analysis is built through explicit transformations.	Explains why maps and metrics are reproducible.
Evaluation posture	Known limits are surfaced rather than hidden.	Prevents unsupported model claims.
Publication surface	Results are communicated with non-claims and bounded scope.	Protects against warning-system confusion.

02 / GEOGRAPHIC INTERPRETATION

Maps as executive evidence, not decoration

The maps help a reader understand concentration patterns, regional separation, and tectonic framing without converting the prototype into an official geophysical product.

OBSERVE Pacific-margin and Gulf-of-California concentration become visually clear, while event-level evidence and regional summaries remain separated.

DECISION VALUE

The visuals justify regional aggregation instead of treating Mexico as a uniform analytical surface.

QUALITY SIGNAL

Clean tooltip fields and separated map layers protect publication credibility.

Boundary: these maps support descriptive prototype reading only; they are not official hazard maps or authoritative plate-boundary products.

Mexico Executive Map

Total Curated Events

43,269

Coverage Start

1984-01-02

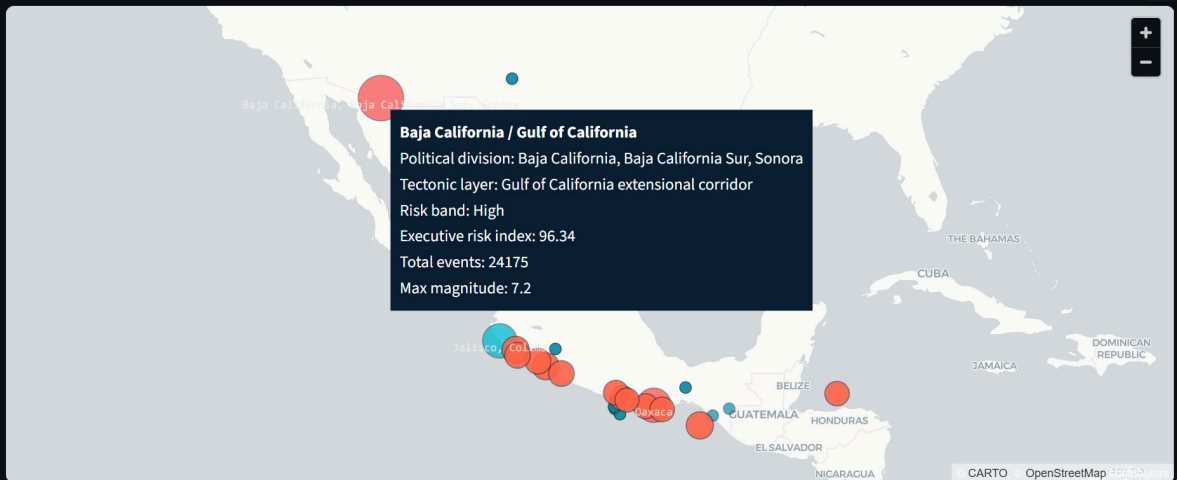
Coverage End

2026-04-11

Strongest Magnitude

8.20

This map presents a **probabilistic-descriptive national view** of the current repository dataset. It helps the reader interpret concentration, stronger events, prototype-region prominence, and bounded AI/ML-oriented analytical reading without implying official hazard or warning claims.



Executive map: concentration, selected regional prominence, and normalized tooltip semantics.

Complementary Context Maps

These supporting maps are intentionally separated to avoid visual crowding and to distinguish event-level evidence from tectonic context.

Complementary Recorded Events Map

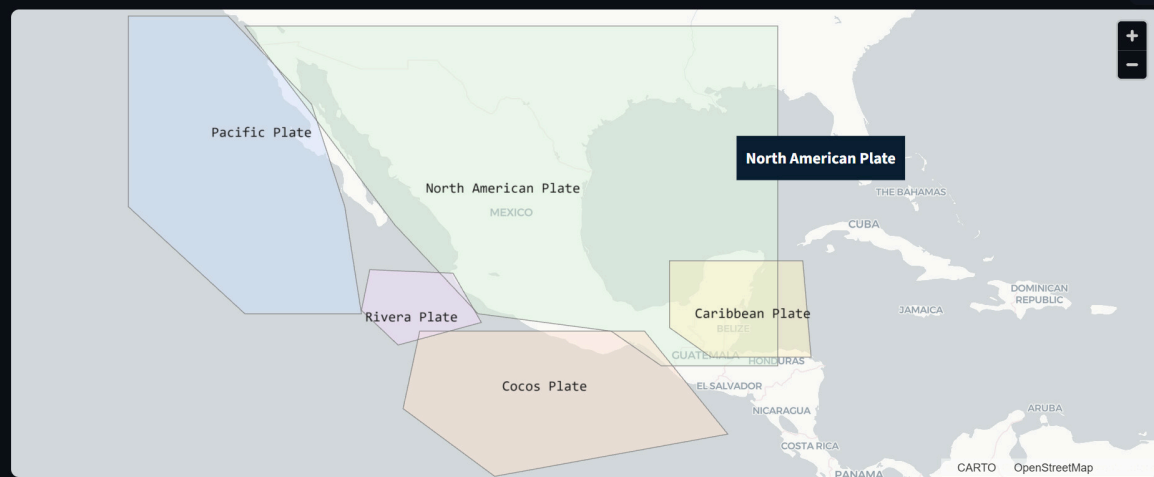
This map complements the executive circles by showing recorded event-level distribution and magnitude in a cleaner standalone view.



Recorded events: supporting event-level evidence, intentionally separated from aggregate regional reading.

Tectonic Plate Context Schematic

This schematic shows approximate relationship, position, and coexistence of the main tectonic plates relevant to Mexico. It is a contextual engineering aid, not an authoritative geophysical boundary product.



Tectonic context: an engineering aid for interpretation, not a formal boundary authority.

03 / REGIONAL INTERPRETATION

From national pattern to selected-region reading

The regional view is the bridge between map impression and defensible analytical explanation. It turns national concentration into one selected prototype band without presenting it as official zoning.

OBSERVE

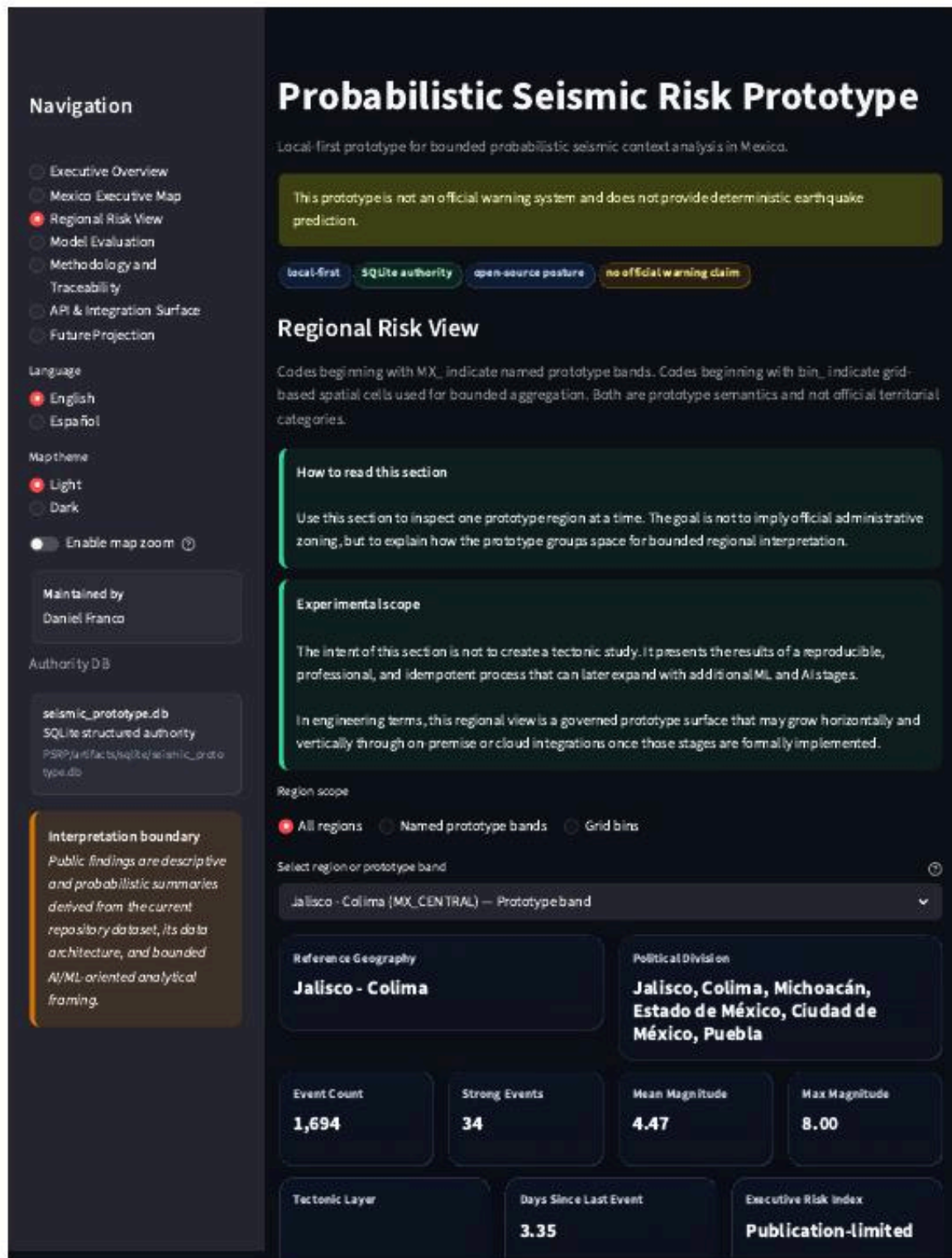
Reference geography, political division, magnitude statistics, event count, and risk index are read as a regional summary.

IMPLICATION

This supports a portfolio narrative around data modeling, regional aggregation, and bounded AI/ML-oriented communication.

BOUNDARY

Prototype bands are semantic aggregation units; they do not replace civil-protection or scientific zoning.



Content-only capture avoids stale sidebar details and focuses the reader on the analytical drill-down.

Bounded evaluation is stronger than unsupported polish

The evaluation section is valuable because it states what is available, what is limited, and what the current model posture can support.

OBSERVE

Publication-limited badges and evaluation cards make constraints visible instead of hiding them behind confident language.

DECISION VALUE

The project shows familiarity with model evaluation surfaces while maintaining claim discipline.

BOUNDARY

Evaluation supports traceability and communication maturity; it is not an earthquake-prediction claim.

Navigation

- Executive Overview
- Mexico Executive Map
- Regional Risk View
- Model Evaluation
- Methodology and Traceability
- API & Integration Surface
- Future Projection

Language

- English
- Español

Map theme

- Light
- Dark

☐ Enable map zoom ⓘ

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Daniel Franco

Authority DB

```
seismic_prototype.db
SQLite structured authority
PSRP/artifacts/sqlite/seismic_prototype.db
```

Interpretation boundary

Public findings are descriptive and probabilistic summaries derived from the current repository dataset, its data architecture, and bounded AI/ML-oriented analytical framing.

Probabilistic Seismic Risk Prototype

Local-first prototype for bounded probabilistic seismic context analysis in Mexico.

This prototype is not an official warning system and does not provide deterministic earthquake prediction.

local-first
SQLite authority
open-source posture
no official warning claim

Model Evaluation

Prototype Objective and Evaluation Scope

The objective here is not to publish a scientific earthquake study. The prototype demonstrates a local-first data architecture, bounded probabilistic aggregation, and a governed publication surface for communicating regional seismic context.

- It is a prototype for probabilistic regional risk communication, not an operational warning system.
- It is not a deterministic earthquake prediction engine.
- It is not a formal geophysical study intended to replace hazard maps, official seismological analysis, or civil-protection procedures.
- Its value lies in architecture, data traceability, bounded analytics, and interpretable publication output.

Engineering interpretation

From an engineering perspective, this section demonstrates governance discipline: the system only elevates what is persisted, traceable, and presentation safe. Missing evidence is treated as a managed limitation instead of being hidden or replaced by speculation.

Publication badges

baseline_publication_limited
not published in current release
local-first authority
bounded interpretation

Current Methodological Elements

The current publication surface is supported by engineering and analytical choices that remain bounded and auditable.

- SQLite as local structured authority for prototype publication artifacts.
- Regional aggregation and grid/bin abstractions to support local-first spatial summarization.
- Temporal windows and derived features compatible with AI/ML-oriented workflows.
- Persisted reports and plots used as public evidence rather than ad hoc narrative claims.

A bounded model posture protects the repository from overclaiming while preserving analytical value.

API surface as controlled inspection path

The API is best interpreted as a local read-oriented integration surface: useful for inspection, automation, and future extension, but not presented as a production backend.

OBSERVE

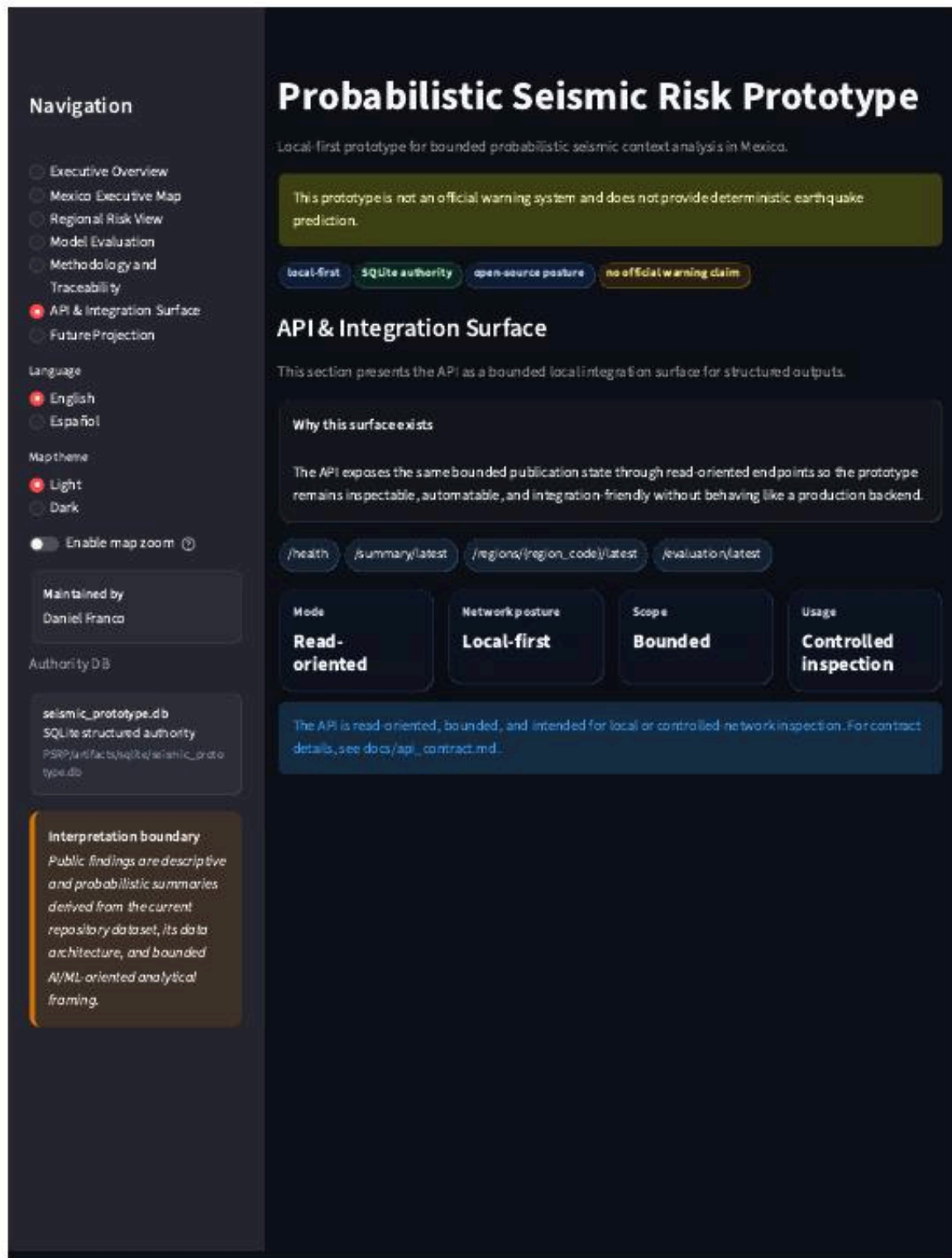
Endpoints expose summary, region, and evaluation payloads as structured read surfaces.

IMPLICATION

The project separates data authority, web presentation, and integration access.

BOUNDARY

LAN/local controlled inspection is current scope; cloud and production operations remain future projection.



The API surface supports review and reproducibility without expanding operational claims.

Evidence-based publication conclusion

PSRP demonstrates a bounded local-first architecture for probabilistic regional seismic-risk context: 43,269 curated events, 42.27 years of coverage, strongest magnitude 8.20, 179 strong events at magnitude 6.0 or higher, and a 30-day feature window organized through a SQLite authority surface.

EVIDENCE BASE	The visible dataset summary supports national-scale reading while preserving a clear distinction between curated historical events and derived regional interpretation.
TECHNICAL VALUE	The strongest professional contribution is traceability: raw evidence, SQLite authority, derived features, regional aggregation, model posture, and publication-safe presentation remain separated.
BOUNDARY	The prototype communicates probabilistic context only; it does not claim official warning capability, deterministic earthquake prediction, or production operational maturity.

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local-first SQLite authority open-source posture no official warning claim

Future Projection

This section describes the next engineering stages that could expand the prototype toward broader ML/AI integration without breaking its current local-first and governed design.

Strategic expansion posture
 The current prototype is intentionally bounded. Its future value comes from disciplined expansion: more capable ML/AI stages, richer geospatial authority, and optional on-premise or cloud integration that preserves traceability, idempotence, and architectural coherence.

Why this matters next
 The current repository already demonstrates bounded value. The next phase should not chase novelty first; it should extend the same governance, traceability, and architectural discipline into richer AI/ML and geospatial capability.

Integration posture
 Future integration should remain optional and layered: on-premise services for low-latency local experimentation, cloud augmentation for heavier workloads, and canonical geometry only when formally ingested and validated.

On-premise AI services
 Deploy local inference or feature-ranking services near the SQLite authority to support low-latency experimentation with bounded MLOps.

Cloud augmentation
 Add optional cloud pipelines for heavier training, archival replication, or federation without making the prototype cloud-dependent.

Canonical geometry ingestion
 Formally ingest political boundaries, tectonic geometries, and validated geospatial layers before presenting them as authoritative map overlays.

Expanded ML stages
 Introduce richer modeling stages such as calibration, uncertainty monitoring, temporal drift checks, and scenario analysis under explicit governance gates.

Roadmap Summary

Roadmap interpretation: ambitious but explicitly controlled.

Executive close

The visual review index concludes that the prototype's publication value comes from disciplined evidence handling: a current curated dataset, local SQLite authority, interpretable regional aggregation, bounded evaluation posture, and explicit separation from emergency-warning or deterministic-prediction claims.

STRENGTH

Traceable local-first architecture with structured authority and derived publication artifacts.

CONSTRAINT

Prototype-grade findings only; no official warning, deterministic prediction, or production operational maturity.

CONTINUITY

Future ML, canonical geometry, larger datasets, and optional cloud augmentation remain roadmap extensions, not current dependencies.

PSRP Executive Controlled Review - interpretive executive report for the visible prototype surface. It summarizes evidence, traceability, and bounded claims without extending operational scope.